State space models II

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ECMA 31150, Spring 2026

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Plan for this slide deck

1. Modeling time-varying parameters
2. Missing data
3. Mixed-frequency data, nowcasting
4. High-dimensional factor models
5. Models with regime switching

State space model with time-varying coefficients

- Consider a state space model where the coefficients depend on time:

$$\begin{aligned} \boldsymbol{y}_t &= \boldsymbol{c}_t + \boldsymbol{G}_t \boldsymbol{s}_t + \boldsymbol{v}_t, \\ \boldsymbol{s}_t &= \boldsymbol{F}_t \boldsymbol{s}_{t-1} + \boldsymbol{u}_t, \\ \begin{pmatrix} \boldsymbol{v}_t \\ \boldsymbol{u}_t \end{pmatrix} &\overset{indep}{\sim} N\left(\boldsymbol{0}, \begin{pmatrix} \boldsymbol{V}_t & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{U}_t \end{pmatrix}\right). \end{aligned}$$

- Implicit assumption: whatever is driving the time-variation in the coefficients is only related to the conditional variances of the shocks, not their conditional means.
  - Rules out that $\boldsymbol{u}_t$ is directly affecting the path for $\boldsymbol{G}_t$, say.
- The Kalman filter and smoother algorithms apply exactly as in the last slide deck. We just need to put time subscripts on all the coefficient vectors/matrices: $\boldsymbol{c} \to \boldsymbol{c}_t$, $\boldsymbol{G} \to \boldsymbol{G}_t$, etc.
  - Smoother algorithm: when defining $\boldsymbol{J}_t$, substitute $\boldsymbol{F} \to \boldsymbol{F}_{t+1}$.

Example: time-varying parameters

- Economic relationships may change over time.
- Consider a time series regression model with time-varying slope:

$$
\begin{aligned}
y_t &= \beta_t x_t + v_t, \\
\beta_t &= \beta_{t-1} + u_t, \\
\begin{pmatrix} v_t \\ u_t \end{pmatrix} &\overset{iid}{\sim} N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_v^2 & 0 \\ 0 & \sigma_u^2 \end{pmatrix} \right).
\end{aligned}
$$

- Fits into the framework on the previous slide:

$$
\boldsymbol{s}_t \equiv \beta_t, \quad \boldsymbol{c}_t \equiv 0, \quad \boldsymbol{G}_t \equiv x_t, \quad \boldsymbol{F}_t \equiv 1, \quad \boldsymbol{V}_t \equiv \sigma_v^2, \quad \boldsymbol{U}_t \equiv \sigma_u^2.
$$

- Note: we're assuming that $\{x_t\}$ is independent of the shocks $\{u_t, v_t\}$.
- Exercise: propose a more general state space model where the regression residual is serially correlated (e.g., an AR(1) process).

```
def kalman_filter(Y, c, G, V, F, U, s0, P0):
    T, n = Y.shape
    m = F.shape[-1]

    s_pred = np.zeros((T, m))       # s_{t|t-1}
    P_pred = np.zeros((T, m, m))    # P_{t|t-1}
    s_filt = np.zeros((T, m))       # s_{t|t}
    P_filt = np.zeros((T, m, m))    # P_{t|t}
    y_pred = np.zeros((T, n))       # y_{t|t-1}
    Q_pred = np.zeros((T, n, n))    # Q_{t|t-1}

    s_prev, P_prev = s0, P0
    for t in range(T):
        # Predict
        s_pred[t] = F[t] @ s_prev
        P_pred[t] = F[t] @ P_prev @ F[t].T + U[t]
        y_pred[t] = c[t] + G[t] @ s_pred[t]
        Q_pred[t] = G[t] @ P_pred[t] @ G[t].T + V[t]

        # Gain
        K = np.linalg.solve(Q_pred[t], G[t] @ P_pred[t]).T

        # Update
        innov = Y[t] - y_pred[t]
        s_filt[t] = s_pred[t] + K @ innov
        P_filt[t] = P_pred[t] - K @ Q_pred[t] @ K.T

        s_prev, P_prev = s_filt[t], P_filt[t]

    return {'s_pred': s_pred, 'P_pred': P_pred,
            's_filt': s_filt, 'P_filt': P_filt,
            'y_pred': y_pred, 'Q_pred': Q_pred}
```

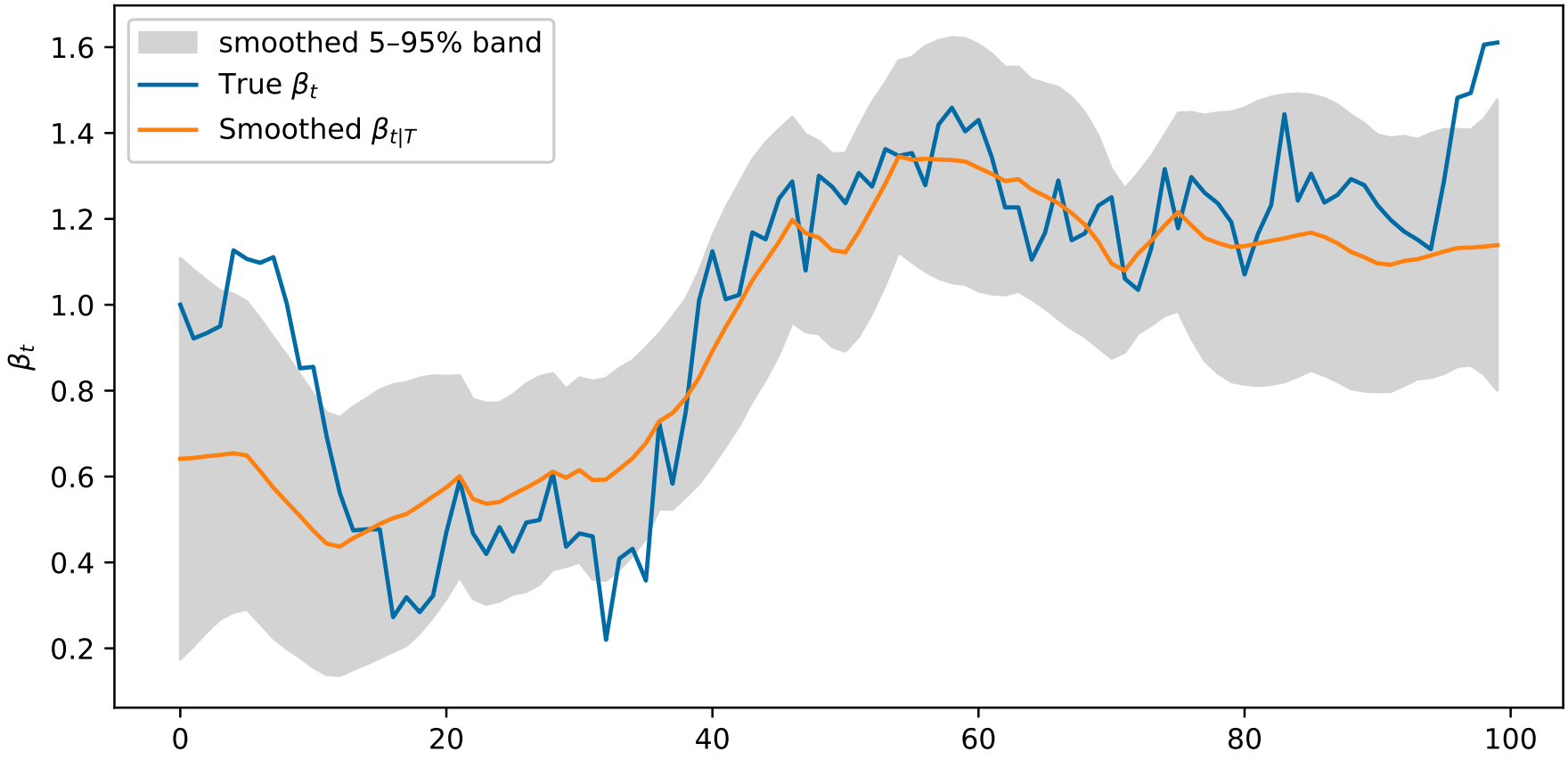
```
def kalman_smoother(filt, F):
    s_pred, P_pred = filt['s_pred'], filt['P_pred']
    s_filt, P_filt = filt['s_filt'], filt['P_filt']
    T, m = s_filt.shape

    s_smooth = np.zeros((T, m))     # s_{t|T}
    P_smooth = np.zeros((T, m, m))  # P_{t|T}

    # Initialize at final filtered state
    s_smooth[T - 1] = s_filt[T - 1]
    P_smooth[T - 1] = P_filt[T - 1]

    for t in range(T - 2, -1, -1):
        J = np.linalg.solve(P_pred[t + 1], F[t + 1] @ P_filt[t]).T
        s_smooth[t] = s_filt[t] + J @ (s_smooth[t + 1] - s_pred[t + 1])
        P_smooth[t] = P_filt[t] + J @ (P_smooth[t + 1] - P_pred[t + 1]) @ J.T

    return {'s_smooth': s_smooth, 'P_smooth': P_smooth}
```



$$y_t = \beta_t x_t + v_t, \quad \beta_t = \beta_{t-1} + u_t, \quad \beta_0 = 1$$

$$\sigma_u = 0.1, \quad \sigma_v = 0.5, \quad T = 100$$

$$x_t \sim \text{AR}(1) \text{ with } \rho = 0.7 \text{ and } \text{Var}(x_t) = 1$$

Missing data

- State space models are great at dealing with missing data.
- Consider a constant-coefficient state space model:

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{G}\boldsymbol{s}_t + \boldsymbol{v}_t, \quad \boldsymbol{v}_t \overset{iid}{\sim} N(\boldsymbol{0}, \boldsymbol{V}),$$
$$\boldsymbol{s}_t = \boldsymbol{F}\boldsymbol{s}_{t-1} + \boldsymbol{u}_t, \quad \boldsymbol{u}_t \overset{iid}{\sim} N(\boldsymbol{0}, \boldsymbol{U}).$$

- But suppose that at some time points $t \in \mathcal{T}_{\text{missing}}$, the first variable y_{1t} is **missing**: we only observe $y_{2t}, \ldots, y_{nt}$ at these points in time.
- We will see that a model with time-varying coefficients can capture the missing data.

Nov. 19, 2025 12:57pm ET

BLS Says It Won't Release an October Jobs Report

By Matt Grossman, Reporter

The Bureau of Labor Statistics said Wednesday that it won't publish a standalone October jobs report.

BLS said it plans to release the November jobs report on Tuesday Dec. 16. October payrolls data will appear in the November jobs report. BLS won't publish an October unemployment rate.

The BLS and the other U.S. government statistics agencies are getting back up to speed after the six-week shutdown hampered their operations for all of October into this month. A delayed September jobs report, which was almost camera ready before the shutdown began, is on the way Thursday morning.

https://www.wsj.com/economy/bls-wont-publish-october-inflation-report-dcf45b03

Missing data: state space model

- Define

$$\boldsymbol{c}_t \equiv \begin{cases} \boldsymbol{c} & \text{for } t \notin \mathcal{T}_{\text{missing}}, \\ \boldsymbol{c} \text{ with first element replaced by zero} & \text{for } t \in \mathcal{T}_{\text{missing}}, \end{cases}$$

$$\boldsymbol{G}_t \equiv \begin{cases} \boldsymbol{G} & \text{for } t \notin \mathcal{T}_{\text{missing}}, \\ \boldsymbol{G} \text{ with first row replaced by zeros} & \text{for } t \in \mathcal{T}_{\text{missing}}, \end{cases}$$

$$\boldsymbol{V}_t \equiv \begin{cases} \boldsymbol{V} & \text{for } t \notin \mathcal{T}_{\text{missing}}, \\ \begin{pmatrix} \omega^2 & \boldsymbol{0}_{1\times(n-1)} \\ \boldsymbol{0}_{(n-1)\times 1} & \boldsymbol{V}_{2:,2:} \end{pmatrix} & \text{for } t \in \mathcal{T}_{\text{missing}}, \end{cases}$$

  where $\boldsymbol{V}_{2:,2:}$ contains rows 2–n and columns 2–n of $\boldsymbol{V}$, and ω^2 is an arbitrary number.

- Then consider the state space model with time-varying coefficients

$$\boldsymbol{y}_t = \boldsymbol{c}_t + \boldsymbol{G}_t \boldsymbol{s}_t + \boldsymbol{v}_t, \quad \boldsymbol{v}_t \sim N(\boldsymbol{0}, \boldsymbol{V}_t),$$
$$\boldsymbol{s}_t = \boldsymbol{F}\boldsymbol{s}_{t-1} + \boldsymbol{u}_t, \quad \boldsymbol{u}_t \sim N(\boldsymbol{0}, \boldsymbol{U}).$$

Missing data: state space model (cont.)

$$\boldsymbol{y}_t = \boldsymbol{c}_t + \boldsymbol{G}_t \boldsymbol{s}_t + \boldsymbol{v}_t, \quad \boldsymbol{v}_t \sim N(\boldsymbol{0}, \boldsymbol{V}_t),$$
$$\boldsymbol{s}_t = \boldsymbol{F}\boldsymbol{s}_{t-1} + \boldsymbol{u}_t, \quad \boldsymbol{u}_t \sim N(\boldsymbol{0}, \boldsymbol{U})$$

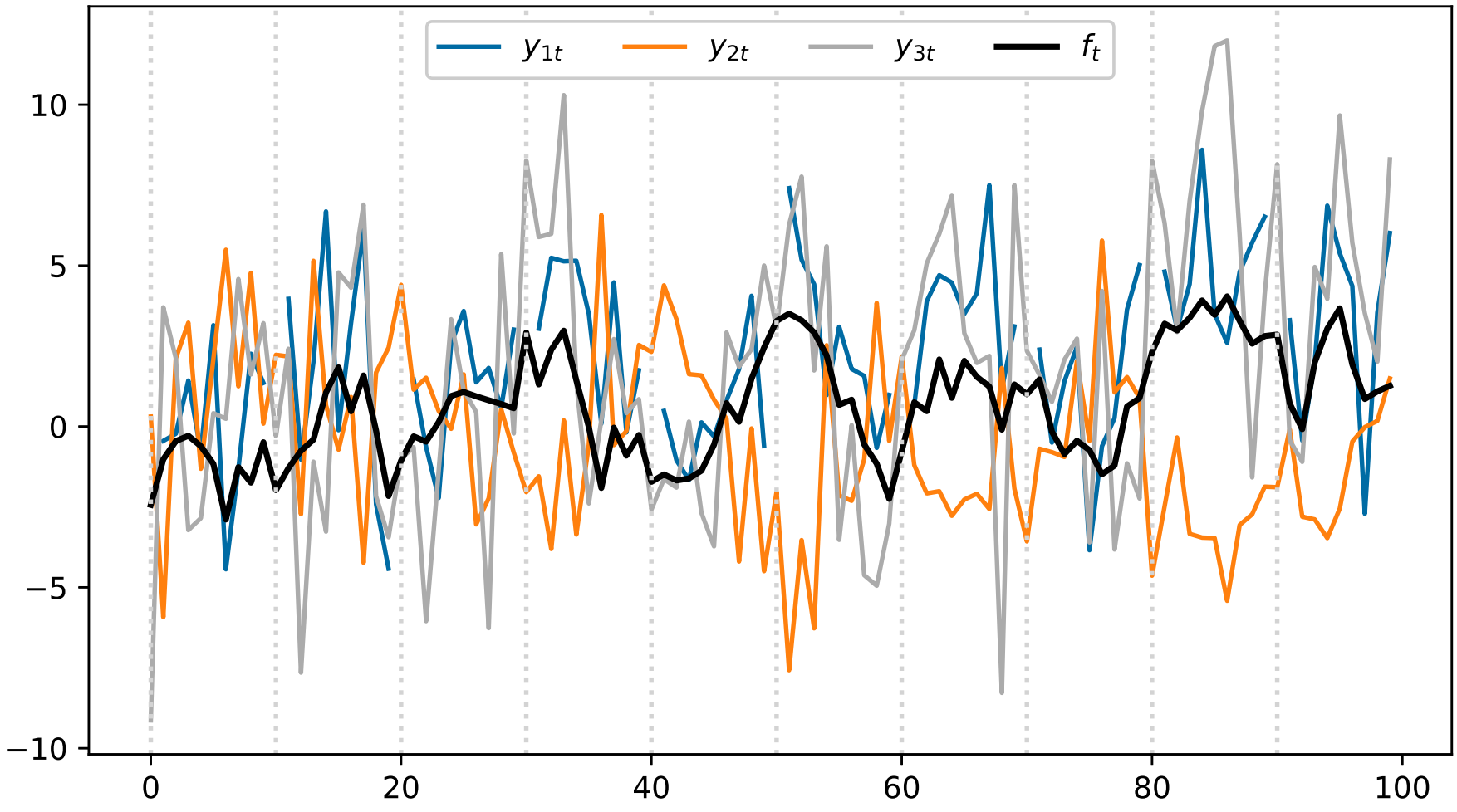
- At dates $t \notin \mathcal{T}_{\text{missing}}$, the model has $\boldsymbol{c}_t = \boldsymbol{c}$, $\boldsymbol{G}_t = \boldsymbol{G}$, $\boldsymbol{V}_t = \boldsymbol{V}$.
- At dates $t \in \mathcal{T}_{\text{missing}}$, the observation equation for y_{1t} becomes

  $$y_{1t} = v_{1t}, \quad v_{1t} \sim N(0, \omega^2), \quad v_{1t} \text{ is independent of } (v_{2t}, \ldots, v_{nt}).$$

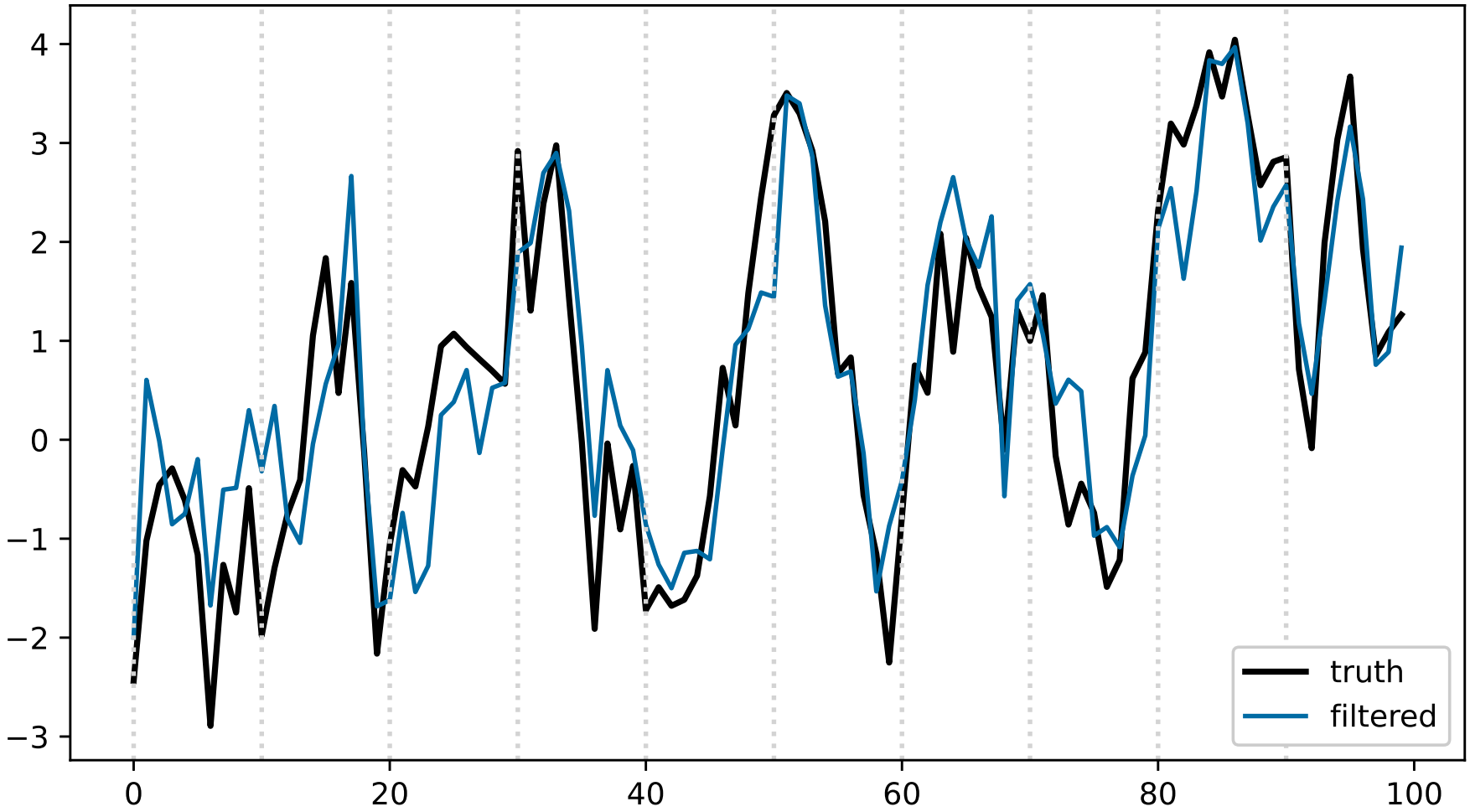
  Therefore, y_{1t} is independent of $\boldsymbol{s}_t$ and of $(y_{2t}, \ldots, y_{nt})$.

Missing data: filter and smoother

- Consider running the Kalman filter/smoother for the model with time-varying coefficients on the previous slide.
- Feed in any arbitrary "data" y_{1t} when it's missing. For example, just set $y_{1t} \equiv 0$ for $t \in \mathcal{T}_{\text{missing}}$.
- By design, the filter/smoother won't use the fake data $\{y_{1t}\}_{t \in \mathcal{T}_{\text{missing}}}$ to infer $\boldsymbol{s}_t$ or to contribute to the likelihood for any parameters.
- But $\boldsymbol{s}_{t|t}$ and the likelihood contribution are still optimally computed based on the available real data $(y_{2t}, \ldots, y_{nt})$ also for $t \in \mathcal{T}_{\text{missing}}$.
- Exercise: how can you generalize the procedure to allow multiple time series to be missing?



Simulation: same dynamic factor model as in previous slide deck, but y_{1t} is missing every 10 periods.



Note: filtered state $s_{t|t}$ is updated also on dates where y_{1t} is missing, using info from (y_{2t}, y_{3t}) and from dynamic model for s_t.

Mixed-frequency data

- A special case of missing data is when we observe time series at multiple frequencies, e.g., monthly and quarterly.
- Can specify a state space model where latent state vector is defined at highest frequency. Low-frequency data is missing in the observation equation in between low-frequency data points.
- Example:
  - Have data on monthly industrial production (IP) and quarterly GDP.
  - State s_t: latent measure of real total output, monthly.
  - IP enters in observation equation every period.
  - GDP only enters observation equation in March, June, September, December, and is related to $s_t + s_{t-1} + s_{t-2}$.

Mixed-frequency data: nowcasting

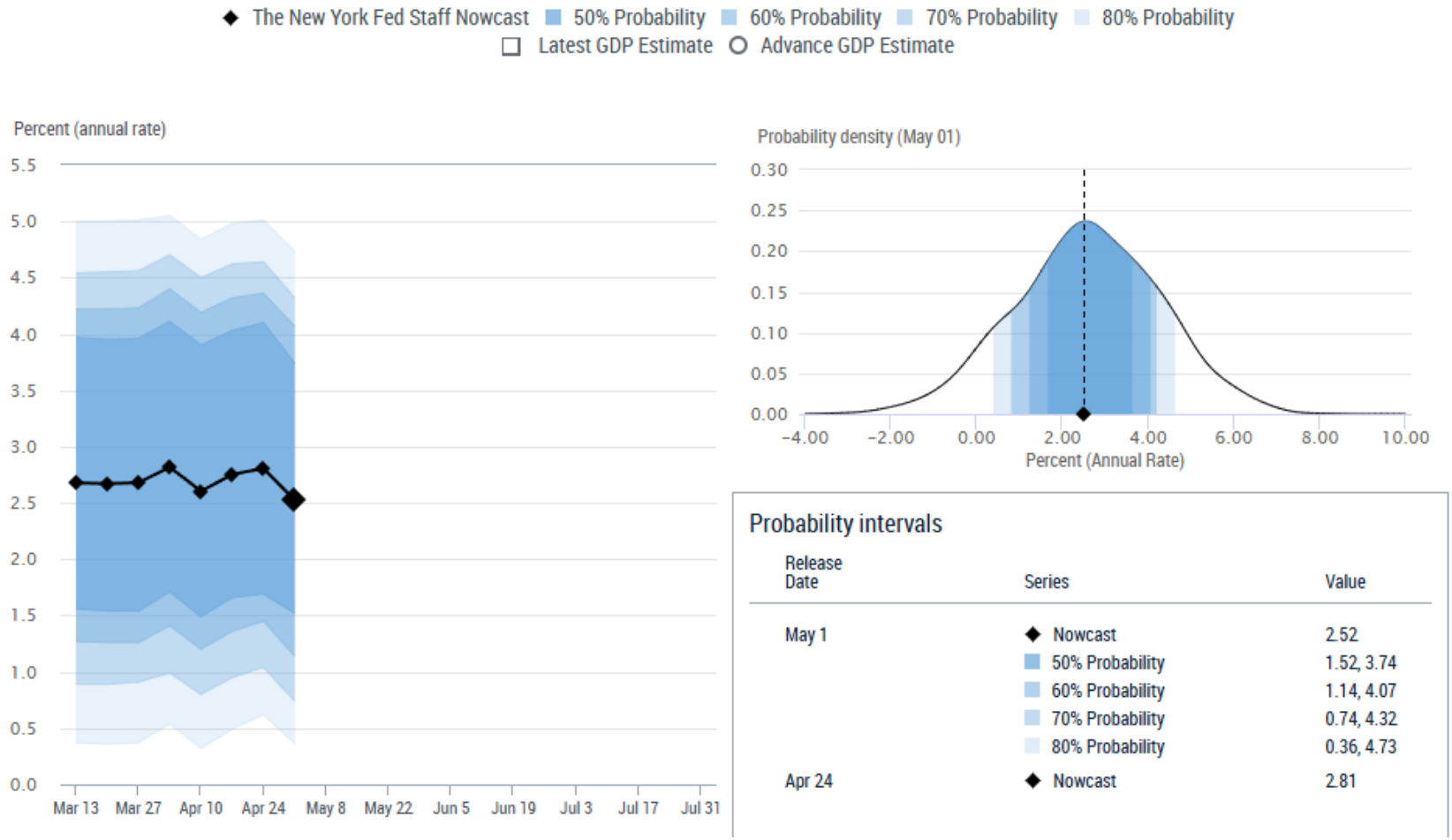
- **Nowcasting** is the act of forecasting the data (e.g., GDP) that applies to the *current* time period *before* the statistical agency actually releases said data.
- We would like to condition the nowcast on a diverse mix of leading indicators measured at various high frequencies, in addition to lower-frequency data (e.g., lagged GDP).
- Data is released by different statistical agencies on different schedules.
- Massive missing data problem at the end of the sample!
- The New York Fed's nowcast model is a sophisticated dynamic factor model that handles the missing/mixed-frequency data.
  - See Bok et al. (2018) and Almuzara et al. (2023) on the reading list.

2026:Q2 | 2026:Q1 | 2025:Q4 | 2025:Q3 Reset to latest nowcast

Latest Release: May 1, 2026

LAYOUT

New York Fed Staff Nowcast



Probability intervals

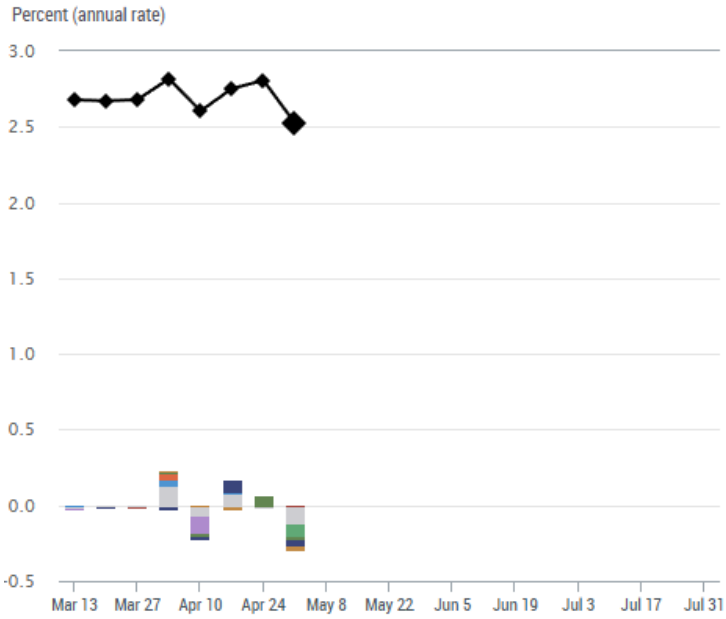
Release Date	Series	Value
May 1	Nowcast	2.52
	50% Probability	1.52, 3.74
	60% Probability	1.14, 4.07
	70% Probability	0.74, 4.32
	80% Probability	0.36, 4.73
Apr 24	Nowcast	2.81

https://www.newyorkfed.org/research/policy/nowcast

Impact of Data Releases

◆ The New York Fed Staff Nowcast ☐ Latest GDP Estimate ○ Advance GDP Estimate

■ Construction ■ Production ■ Surveys ■ Consumption ■ Income ■ Labor ■ Trade ■ Prices ■ Other



Expand

Data Flow (May 1, 2026)

Model update	Release Date	Data Series	Actual	Impact	Nowcast GDP Growth
1-May					2.52
	10:01AM May 01	ISM mfg.: Employment index	46.40	-0.01	
	10:01AM May 01	ISM mfg.: Prices index	84.60	0.01	
	10:01AM May 01	ISM mfg.: PMI composite index	52.70	-0.04	
	8:33AM Apr 30	Real gross domestic product	1.99	-0.07	
	8:31AM Apr 30	PCE less food and energy: Chain price index	0.29	0.00	
	8:31AM Apr 30	PCE: Chain price index	0.66	0.00	
	8:31AM Apr 30	Real disposable personal income	-0.06	-0.01	
	8:34AM Apr 30	Real personal consumption expenditures	0.24	-0.02	
	8:30AM Apr 29	Building permits	-166.00	-0.02	
	8:30AM Apr 29	Housing starts	10.77	0.03	
	8:31AM Apr 29	Manufacturers' inventories: Durable goods	0.24	0.00	
	8:31AM Apr 29	Manufacturers' unfilled orders: All industries	0.11	0.00	
	8:31AM Apr 29	Manufacturers' shipments: Durable goods	0.69	0.00	
	8:37AM Apr 29	Total business inventories	0.90	-0.01	
	8:35AM Apr 29	Merchant wholesalers: Total inventories	1.36	-0.01	
	8:31AM Apr 29	Manufacturers' new orders: Durable goods	0.83	-0.01	
	8:30AM Apr 29	Building permits	152.00	0.01	
	8:30AM Apr 29	Housing starts	-3.00	-0.01	
		Data revisions		0.01	
		Parameter revisions		-0.13	
24-Apr					2.81

High-dimensional factor models

- Consider a dynamic factor model with multiple latent factors and their lags:

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{\lambda}_0 \boldsymbol{f}_t + \boldsymbol{\lambda}_1 \boldsymbol{f}_{t-1} + \cdots + \boldsymbol{\lambda}_p \boldsymbol{f}_{t-p} + \boldsymbol{v}_t.$$

- We assume the dimension r of the latent factors $\boldsymbol{f}_t$ is relatively small.
- But suppose we want to condition our analysis on a large number n of observed variables $\boldsymbol{y}_t = (y_{1t}, \ldots, y_{nt})'$.
  - E.g., data for many sectors/countries.
- We could treat this as a state space model and do Bayesian inference via the Kalman filter likelihood. However, this could be computationally tricky when n is large.

High-dimensional factor models: PCA

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{\lambda}_0 \boldsymbol{f}_t + \cdots + \boldsymbol{\lambda}_p \boldsymbol{f}_{t-p} + \boldsymbol{v}_t.$$

- Turns out we can consistently estimate the factors $\boldsymbol{f}_t$ and loadings $\boldsymbol{\lambda}_j$ (up to some normalizations) using the technique of **principal component analysis** (PCA).
- This involves computing eigenvectors of the $n \times n$ sample variance-covariance matrix of the data $\boldsymbol{y}_t$. Computationally very simple!
  - The PCA estimates could be used as a starting point for subsequent Bayesian inference using state space methods.
- Details: Stock & Watson (2019) textbook, chapter 17.6; Stock & Watson (2016) handbook chapter (on syllabus).
- High-dimensional factor models are some of the most empirically successful economic forecasting models.

Regime switching

- In the linear-Gaussian state space model, the latent state vector $\boldsymbol{s}_t$ is normally distributed.
- In some applications, it makes more sense to think of a *discrete* latent state, also known as a **regime** (e.g., recession vs. expansion).

Regime switching: example

- Regime-switching AR(1) model:

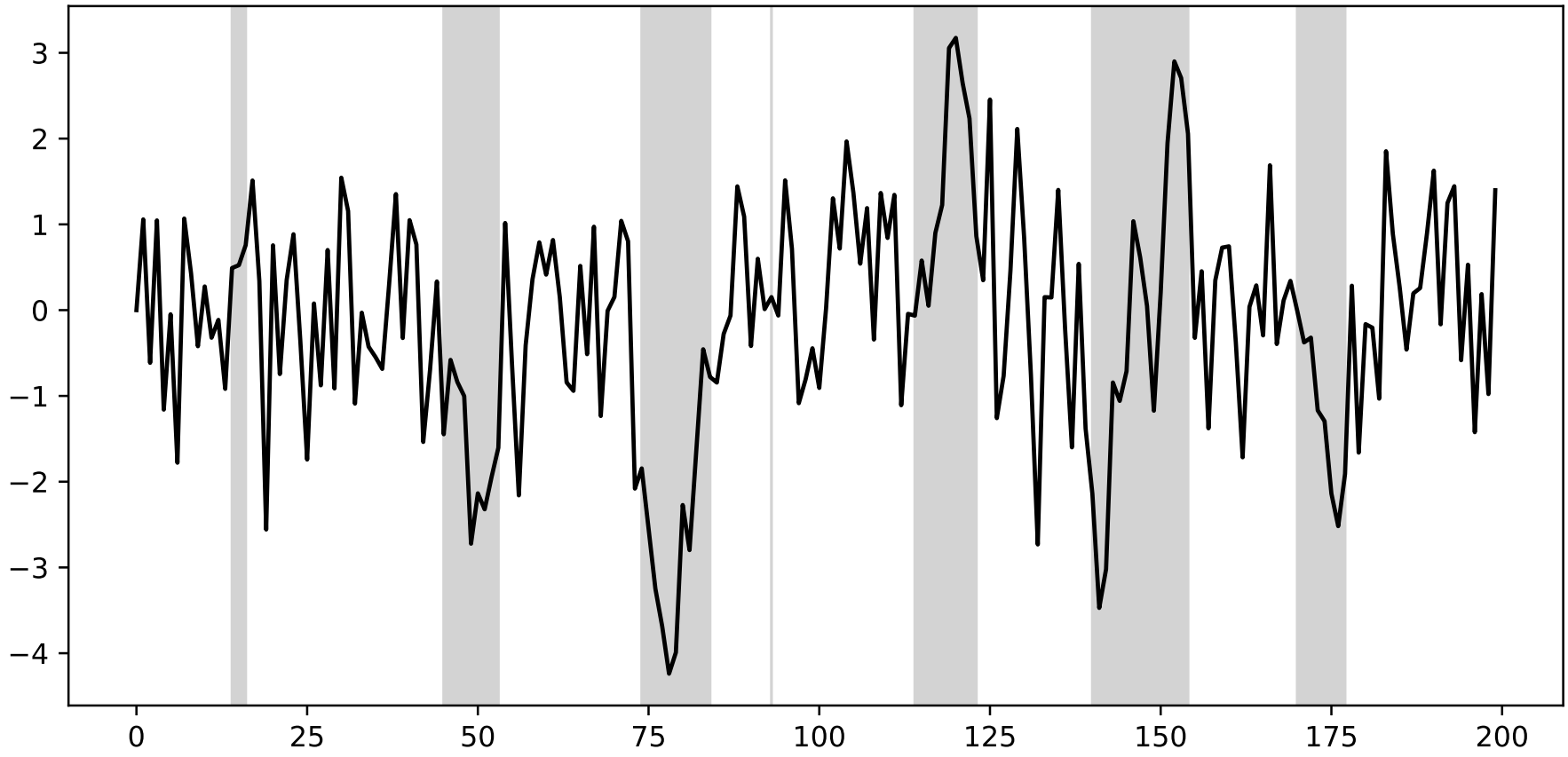
$$y_t = \begin{cases} c_1 + \rho_1 y_{t-1} + u_t, & u_t \sim N(0, \sigma_1^2) \quad \text{when } s_t = 1, \\ c_2 + \rho_2 y_{t-1} + u_t, & u_t \sim N(0, \sigma_2^2) \quad \text{when } s_t = 2. \end{cases}$$

- The latent regime $s_t \in \{1, 2\}$ follows a discrete Markov chain:

$$P(s_t = i \mid s_{t-1} = j, \{y_{t-\ell}, s_{t-1-\ell}\}_{\ell \geq 1}) = P(s_t = i \mid s_{t-1} = j) \equiv \Pi_{ij},$$

  where Π_{ij} are the **transition probabilities**.

- The AR dynamics depend on the regime. The regime switches occasionally according to the transition probabilities.



Simulated regime-switching AR(1) model

Regime 1: $\rho_1 = 0$, regime 2 (in gray): $\rho_2 = 0.9$

Regime s_t switches with probability 0.1 every period

Regime switching: filtering and smoothing

- There exist analogues of the Kalman filter and smoother for models with discrete latent regimes.
  - Can be applied to much more general models than the example on the previous slide.
- The filtering algorithm also produces the likelihood function, so we can do Bayesian inference on the parameters.
- See Hamilton (1994) textbook, chapter 22.

Further reading (on syllabus)

- More examples of state space models + computational tricks:
  - Durbin & Koopman (2012) book.
  - Brockwell & Davis (2016) textbook, chapter 9.
- How to use state space techniques to estimate structural macro models:
  - Herbst & Schorfheide (2015) book.
  - Mancini Griffoli (2013) user guide for Dynare software package.

What did we learn?

- State space model with time-varying coefficients
  - Time-varying parameter models
  - Missing data
  - Mixed-frequency data, nowcasting
- High-dimensional factor models
- Discrete regime switching

Looking ahead

- How can we model data that is not stationary?
- How do our usual statistical procedures (e.g., regression) perform when we apply them to non-stationary data?
- How can we test the stationarity assumption?